

Akash Shivaji Varude

M.Tech, Computer Science and Engineering
Indian Institute of Technology Kanpur

✉ akashvarude305@gmail.com / ☎ 9595609094
🌐 LinkedIn / 🐙 GitHub / 📁 Portfolio

PROFESSIONAL EXPERIENCE

Associate Developer | SAP Labs India

(Jul'25 – Present)

- Driving **CI/CD automation for DPQuery URM** covering automated environment deployments, validation and Level-0 tenant provisioning tests; also integrated deployment safeguards and lifecycle controls to make URM changes more repeatable and reduce reliance on manual deployment steps.
- Worked on end-to-end **tenant provisioning validation** by moving smoke-test execution toward dedicated CI identities and controlled DPQuery environments, validating resource creation, readiness and cleanup as part of the deployment pipeline so provisioning regressions can be detected before changes progress further.
- Enhanced the DPQuery **SPII Handler** with **SAP BTP Feature Flags** and feature-controlled **BDC Connect routing**, enabling runtime tenant-mapping changes without redeployments; also added token caching and automated tests.
- Extended **DDL support in HANA for Delta virtual objects** for CREATE/ALTER execution by coordinating changes with remote Spark controllers; also designed **compensation and rollback handling** for distributed metadata operations to preserve consistency and revert partially completed operations when failures occur.

Senior Student Research Associate | Center for Developing Intelligent Systems, IIT Kanpur

(Mar'24 – May'25)

M.Tech Thesis: Towards Revocable Biometrics **Advisor:** Prof. Nisheeth Srivastava

- Developed an end-to-end **signature verification and forgery-detection system** covering data collection, deep-learning model training, backend inference and mobile-based verification.
- Collected signature data from **30 subjects** and constructed **3,000 static image pairs** and **4,350 dynamic stroke pairs** for Siamese-network training and evaluation. Developed an image-based **Siamese neural network** for learning spatial signature embeddings from static signature images.
- Developed a dynamic-signature **BiLSTM with Attention Siamese network** using temporal characteristics including pressure, velocity, timing and acceleration. Achieved **99.11% accuracy** with the image-based model and **99.84% accuracy** with the stroke-based model; the stroke model achieved **zero false positives** on the test set.
- Trained the image-based model using **contrastive loss** and the stroke-based model using a **weighted angular-margin loss** for learning discriminative signature representations.
- Designed a **minutiae-extraction algorithm** for identifying fingerprint termination and bifurcation points. Implemented **minutiae-cylinder-based feature representations** and probabilistic matching for fingerprint comparison. Evaluated the verification pipeline on **5,112 fingerprint pairs** from the **FVC2000 dataset**, achieving **96% accuracy**.

Research Associate | Indian Institute of Technology Hyderabad

(Jul'22 – Dec'22)

- Worked on **continual learning for computer vision**, focusing on image-based continual learning.

Software Engineer | Apisero Global Integration Solutions Pvt Ltd

(Jan'21-June'21)

EDUCATION

Indian Institute of Technology Kanpur

2023 – 2025

M.Tech, Computer Science and Engineering

CPI: 9.14/10

MIT Academy of Engineering, Pune

2017 – 2021

B.Tech, Computer Science and Engineering

CPI: 9.64/10

PUBLICATION

Classification of Online Pernicious Comments using Machine Learning

(Aug'19-Sept'19)

International Journal for Scientific Research and Development (IJSRD) Volume : 7, Issue : 8, Date: 1/11/2019

- Created a **multi-label classification** model using Logistic Regression, SVM, KNN and Decision Trees to detect and categorize toxic comments into six hate categories using a Kaggle dataset of **153,165 labeled comments** for training and evaluation

KEY PROJECTS

Computational Linguistics for Indian Languages | Course Projects: CS689

(Jan'24-Apr'24)

NLP, PyTorch, Hugging Face Transformers, scikit-learn, Text Classification, Machine Translation, Tokenization, Feature Engineering

- MaHate: Marathi Hate Speech Classification:** Developed a **custom architecture** using MahaRoBERTa as the base model and incorporated additional features, outperforming existing models including MahaBERT and XLM-RoBERTa by L3cube
- Trained and tested model on total **25,000 tweets** achieving **83.35% accuracy** with **83.34% macro-F1** score
- Unicode Correction and Feature Extraction:** Developed a Unicode correction algorithm to standardize text data, extracted syllables, performed unigram-bigram frequency analysis and compared various tokenization methods including Unigram, BPE, mBERT, IndicBERT, and white-space tokenizer, tested the performance on more than **400,000 Marathi sentences**
- Cross-Lingual Machine Translation Analysis:** Evaluated machine translation models Distilled NLLB-200, IndicTrans, and ChatGPT on benchmark dataset for Marathi, Hindi, and English and assessed the performance using BLEU and ROUGE metrics

Image Synthesis using Generative Adversarial Networks | Self Project

(July'24-Aug'24)

Generative Adversarial Networks, PyTorch, TorchVision, NumPy, Matplotlib

- Trained GANs on Fashion MNIST dataset with **70000 images** and then on CelebA dataset with **200k images**
- Optimized GAN training by fine-tuning and employing techniques like **label smoothing** and **feature scaling**

TECHNICAL SKILLS

- Programming Languages/Tools:** C, C++, Python, SQL, Git, LaTeX, Google Colab, JavaScript, HTML
- Libraries/Frameworks:** TensorFlow, Pytorch, Numpy, Pandas, OpenCV, Keras, Matplotlib, Node.js
- Areas of Interest:** AI, Machine Learning, Data Science, ML system design, Backend Systems, Distributed Systems, NLP, GANs

ACHIEVEMENTS

- Honored as a Reliance Foundation **Postgraduate Scholar** 2024, selected among top **100 students across India**
- Achieved **(AIR) 156** in GATE CS 2023 and **(AIR) 605** in GATE CS 2022
- Received the **Academic Excellence Award** and secured **Rank 2 in B.Tech** at MIT Academy of Engineering, Pune